SEXUAL & REPRODUCTIVE HEALTH

PT-141 — the full review

MELANOCORTIN RECEPTOR AGONIST (BREMELANOTIDE)

This is the deeper look behind the short version — the real numbers behind the approved trials, how the approved product and a compounded preparation differ, and what your consultation covers.

What it is

PT-141, also known by its chemical name bremelanotide, is a lab-made, seven-amino-acid cyclic peptide built to mimic a natural hormone called alpha-melanocyte-stimulating hormone (α-MSH). As Vyleesi, it is FDA-approved for one specific population: premenopausal women with hypoactive sexual desire disorder (HSDD), where low desire causes marked distress and isn't better explained by another medical or psychiatric condition, a relationship problem, or a medication. Separately from Vyleesi, the same molecule is available as a compounded preparation, sold under its research name PT-141, used off-label for other patients — notably men, for whom no FDA-approved option currently exists for this purpose.

It is most often described alongside PDE5 inhibitors like sildenafil, but the two work in essentially different places in the body. PDE5 inhibitors act locally, relaxing blood vessels in the penis, and only amplify arousal that is already present. Bremelanotide acts further upstream, on receptors in the brain and central nervous system involved in the desire and arousal pathway itself, independent of local blood flow — which is why it is taken as a single as-needed injection well before activity rather than "on demand," and why its side-effect pattern looks nothing like a PDE5 inhibitor's.

The approved product and the compounded product are not interchangeable. Vyleesi is a single, standardized 1.75 mg subcutaneous auto-injector dose with FDA-reviewed manufacturing controls. Compounded or gray-market "PT-141" — including nasal sprays and higher-dose vials — is not reviewed by FDA for identity, purity, potency, or sterility, and the nasal route it commonly uses is the same route an earlier FDA safety review flagged strongly enough to end that entire development program in 2007.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Low sexual desire in premenopausal women (the approved use)	**FDA-approved**	Two large trials showed a real, statistically significant but modest benefit on desire and distress scores; no significant improvement was seen in satisfying sexual events. Nausea and flushing are common side effects.
Male erectile dysfunction, off-label	**Human trials**	Based on an older, FDA safety-halted program; its largest supporting trial (342 men) carries an unresolved 2023 journal integrity notice.
Male low libido without diagnosed ED	**Limited human data**	The male trials that exist measured erectile response, not a validated low-desire outcome; any libido-specific benefit is extrapolated rather than directly demonstrated.

USED FOR	EVIDENCE	WHAT THAT MEANS
Female use outside the approved population	**Not studied**	The label explicitly excludes postmenopausal women; the pivotal trials enrolled premenopausal women only, and no dedicated trial for other populations was identified.
Higher-dose or intranasal protocols sold online	**Not studied**	This is the specific route an earlier FDA review flagged for a blood-pressure signal strong enough to end the intranasal development program; no modern trial of these doses exists.
General desire or performance enhancement	**Theoretical**	Not tested in any population without a diagnosed condition; extrapolated from the drug's proposed brain mechanism, not a demonstrated human effect.

How it works

Bremelanotide is a melanocortin receptor (MCR) agonist — it nonselectively activates several receptor subtypes, most consequentially MC4R, found on central-nervous-system neurons implicated in animal models of sexual desire and arousal circuits, and MC1R, found on melanocytes, activation of which is the mechanistic explanation for the drug's hyperpigmentation side effect. The FDA label is direct that the underlying story stops there: "The mechanism of action of bremelanotide for the treatment of hypoactive sexual desire disorder is unknown." What is documented is receptor pharmacology and the downstream effects — blood pressure and heart rate changes, nausea, pigmentation — that follow from activating multiple melanocortin receptor subtypes at once, not a validated causal chain explaining exactly how that translates into increased desire in a given patient.

What the research actually looked at

The approved indication rests on two identical Phase 3, randomized, placebo-controlled trials (the RECONNECT program) enrolling 1,247 premenopausal women over 24 weeks. Both co-primary endpoints — a validated desire-domain score and a distress score — showed statistically significant improvement over placebo, though the effect sizes were modest by conventional standards (roughly 0.27 to 0.39 on the standard scale). On the secondary endpoint that arguably matters most to a patient — the number of satisfying sexual events — there was no statistically significant difference from placebo. Nausea occurred in 40% of participants (versus 1.3% on placebo) and flushing in 20% (versus 0.3%); 18% of the treatment group discontinued the trial, 8% specifically for nausea.

The off-label male-use literature is considerably thinner and older, tracing almost entirely to the original 1990s–2000s intranasal PT-141 program that FDA safety concerns ended in 2007–2008, before it reached approval. The most-cited controlled study — 342 men who had not responded to sildenafil — reported a "positive clinical response" in 33.5% of the bremelanotide group versus 8.5% on placebo, but this specific paper carries a formal, unresolved 2023 journal Expression of Concern. No trial has tested the doses or routes sold today, and no dedicated trial exists for postmenopausal women or for general enhancement use in people without a diagnosed condition.

Common misunderstandings

- **“It's basically like Viagra.”** Mechanistically these work in close to opposite places. PDE5 inhibitors act peripherally on blood vessels and only amplify arousal already present; bremelanotide acts centrally, in brain pathways linked to desire, independent of local blood flow — which is also why the side-effect profiles look nothing alike.

- **“It's approved for everyone — women, men, general enhancement.”** The label is narrow and explicit: approved only for premenopausal women with a specific diagnosed form of low desire. It states outright that it is not indicated for postmenopausal women, not indicated for men, and not indicated for sexual performance enhancement.
- **“Nausea and flushing are rare side effects.”** They are the opposite of rare. Nausea occurred in 40% of trial participants (versus 1.3% on placebo) and flushing in 20% (versus 0.3%) — common enough that 18% of the treatment group left the trial early.
- **“The PT-141 sold online or through a compounding pharmacy is the same product, at the same dose, as Vyleesi.”** It generally is not. Vyleesi is a single, standardized, FDA-reviewed dose; compounded or gray-market "PT-141" — including nasal sprays and higher-dose vials — is unregulated for identity, purity, potency, and sterility.

Who it is not for

- Uncontrolled hypertension or known cardiovascular disease — an explicit label contraindication, since bremelanotide causes transient blood pressure and heart rate changes after each dose.
- Pregnancy, or trying to conceive without reliable contraception — animal studies showed fetal harm at exposures well above the maximum recommended human dose.
- Low desire caused by another medical condition, psychiatric condition, relationship problem, or medication — the approved trials specifically excluded these causes.
- Postmenopausal women — excluded from the pivotal trials and explicitly outside the label's indication.
- Repeated or frequent dosing beyond label guidance, particularly for anyone at higher risk of visible pigmentation change.
- Anyone sourcing "PT-141" from an unregulated online vendor.
- More than one dose in 24 hours, or more than about 8 doses a month — both exceed the tested and labeled regimen.

Competitive athletes: The terms "melanocortin," "bremelanotide," "PT-141," and "melanotan" do not appear on the current WADA Prohibited List, and no anti-doping sanction case involving it was identified — likely reflecting that bremelanotide's FDA approval (Vyleesi) keeps it out of WADA's catch-all category for substances with no approval anywhere. This should be read as "not found," not as a guaranteed clearance; a competing athlete should independently confirm current status through Global DRO or their national anti-doping agency, and should know that an unregulated compounded or gray-market vial carries its own contamination risk that could trigger a positive test for an unrelated substance.

Approved product, or a compounded preparation — and the difference matters

This is an FDA-approved medicine. It is available as the manufacturer's own approved product, and — separately — as a compounded preparation. Which one is being offered is always stated plainly.

If the approved product is right, that is what gets prescribed:

- **The manufacturer's pen or vial.** Reviewed by the FDA for safety, effectiveness and quality, exactly as labeled.
- **Filled at a licensed pharmacy.** Dispensed from your prescription like any other approved medicine.

If a compounded preparation is clinically appropriate instead, it is prepared by a licensed 503A compounding pharmacy:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not itself an FDA-approved product — but it is still made under a licensed pharmacy's enforceable standards, not an unregulated website.

Where this stands

There is an approved product, for one specific use. Bremelanotide approved as VYLEESI for acquired generalised hypoactive sexual desire disorder in premenopausal women — a real, statistically significant but modest benefit, with nausea (40%) and flushing (20%) common.

Use outside that group is off-label. That includes men, postmenopausal women, and general enhancement — all resting on an older program FDA safety-halted in 2007, whose largest supporting trial carries an unresolved 2023 journal integrity notice.

An approval for one population, on one outcome measure, is not evidence for another population or a different claim.

Clinical detail

Bremelanotide descends from melanotan II, a 1980s University of Arizona sunless-tanning candidate; an unplanned erection during self-experimentation redirected research toward sexual-dysfunction use.

FDA halted the original intranasal PT-141 development program in 2007 over blood-pressure increases; the subsequently reformulated subcutaneous version, focused on premenopausal HSDD, became Vyleesi, approved June 21, 2019 (NDA 210557).

Vyleesi was only the second female-libido medication FDA has ever approved, after flibanserin (Addyi) in 2015.

Neither "bremelanotide" nor "PT-141" currently appears on FDA's active 503A Bulks List, the nominated-but-withdrawn table, or the Category 2 significant-safety-risk table.

A 2020 FDA warning letter to a compounding pharmacy named "Bremelanotide (PT-141)" directly among substances used outside the exemptions available to 503A compounders.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

PT-141 is FDA-approved for the uses described as approved above. Off-label uses discussed here are not FDA-approved uses. A compounded preparation, where used instead of the approved product, is not itself an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any unapproved purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.